MATH 152
Mrs. Bonny Tighe

QUIZ 4
7.2, 7.3
25 points

Name Solutions

TA_____ Wed 9/26/2018

1. Evaluate the integrals.

4pts a) $\int t\,(\sin^2 t)\;dt$

$u = t \qquad dv = \sin^2 t = \frac{1}{2} - \frac{1}{2}\cos 2t\,dt$

$du = dt \qquad v = \frac{1}{2}t - \frac{1}{4}\sin 2t$

$t\left(\frac{1}{2}t - \frac{1}{2}\sin 2t\right) - \int\left(\frac{1}{2}t - \frac{1}{2}\sin 2t\right)dt$

$\frac{1}{2}t^2 - \frac{1}{4}t\sin 2t - \left[\frac{1}{4}t^2 + \frac{1}{4}\cos 2t\right] + C$

$\frac{1}{2}t^2 - \frac{1}{4}t\sin 2t - \frac{1}{4}t^2 - \frac{1}{8}\cos 2t + C$

$\frac{1}{4}t^2 - \frac{1}{4}t\sin 2t - \frac{1}{8}\cos 2t + C$

4pts b) $\int \frac{\tan^2\sqrt{x}}{\sqrt{x}}\,dx$

$m = \sqrt{x} \qquad dm = \frac{1}{2\sqrt{x}}\,dx \qquad dx = 2m\,dm$

$\int \frac{\tan^2 m}{m}(2m\,dm) = 2\int \tan^2 m\,dm$

$2\int(\sec^2 m - 1)\,dm = 2\tan m - 2m + C$

$2\tan\sqrt{x} - 2\sqrt{x} + C$

4pts c) $\int \cos^3(4x)\,dx$ $= \int \cos^2 4x \cos 4x\,dx$

$\int (1-\sin^2 4x)\cos 4x\,dx$

$u = \sin 4x$

$du = 4\cos 4x\,dx$

$\frac{du}{4\cos 4x} = dx$

$\int (1-u^2)(\cos 4x)\left(\frac{du}{4\cos 4x}\right)$

$\frac{1}{4}\int (1-u^2)\,du = \frac{1}{4}\left(u - \frac{1}{3}u^3\right) + C$

$\frac{1}{4}\sin(4x) - \frac{1}{12}\sin^3(4x) + C$

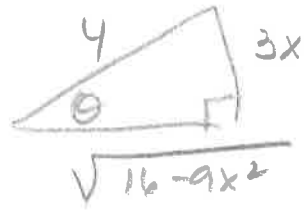
4pts d) $\int \frac{\sqrt{16-9x^2}}{x}\,dx$

$3x = 4\sin\theta$

$3x/4 = \sin\theta$

$x = 4/3 \sin\theta$

$dx = 4/3 \cos\theta\,d\theta$



$\int \frac{\sqrt{16-16\sin^2\theta}}{4/3\sin\theta}\,4/3\cos\theta\,d\theta$

$\int \frac{4\cos\theta\ 4/3\cos\theta\,d\theta}{4/3\sin\theta} = 4\int \frac{\cos^2\theta}{\sin\theta}\,d\theta =$

$4\int \frac{1-\sin^2\theta}{\sin\theta}\,d\theta = 4\int\left(\frac{1}{\sin\theta} - \sin\theta\right)d\theta$

$4\int(\csc\theta - \sin\theta)\,d\theta = 4\ln|\csc\theta - \cot\theta| + 4\cos\theta + C$

$4\ln\left|\frac{4}{3x} - \frac{\sqrt{16-9x^2}}{3x}\right| + 4\left(\frac{\sqrt{16-9x^2}}{4}\right) + C$

5pts

e) $\int_{3\sqrt{2}}^{6} \frac{x^3}{\sqrt{x^2-9}}dx$

$x = 3\sec\theta$

$dx = 3\sec\theta\tan\theta\, d\theta$

$3\sqrt{2} = 3\sec\theta \quad \theta = \pi/4$

$6 = 3\sec\theta \quad \theta = \pi/3$

$$\int_{\pi/4}^{\pi/3} \frac{27\sec^3\theta \; 3\sec\theta\tan\theta\, d\theta}{\sqrt{9\sec^2\theta - 1}} = \int_{\pi/4}^{\pi/3} \frac{81\sec^4\theta\tan\theta\, d\theta}{3\tan\theta}$$

$$27\int_{\pi/4}^{\pi/3} \sec^4\theta\, d\theta = 27\int_{\pi/4}^{\pi/3} \sec^2\theta\sec^2\theta\, d\theta =$$

$$27\int_{\pi/4}^{\pi/3} \sec^2\theta(1+\tan^2\theta)\, d\theta = 27\int_{\pi/4}^{\pi/3} \sec^2\theta\, d\theta + 27\int_{\pi/4}^{\pi/3} \tan^2\theta\sec^2\theta\, d\theta$$

$u^2\, du$

$$27\left[\tan\theta + \tfrac{1}{3}\tan^3\theta\right]_{\pi/4}^{\pi/3} =$$

$$27\left[\tan\pi/3 + 1/3\tan^3\pi/3 - (\tan\pi/4 + \tfrac{1}{3}\tan^3\pi/4)\right]$$

$$27\left[\sqrt{3} + \tfrac{1}{3}(\sqrt{3})^3 - 1 - 1/3\right] = 27\left[\sqrt{3} + \sqrt{3} - 4/3\right]$$

$$27\left[2\sqrt{3} - 4/3\right]$$

4pts

f) $\int \frac{1}{\sqrt{x^2+2x+5}}dx$

$$\int \frac{1}{\sqrt{(x+1)^2+4}}dx$$

$x+1 = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$

$$\int \frac{2\sec^2\theta\, d\theta}{\sqrt{4\tan^2\theta+1}} = \int \frac{2\sec^2\theta\, d\theta}{2\sec\theta} = \int \sec\theta\, d\theta$$

$$\ln|\sec\theta + \tan\theta| + C$$

$$\ln\left|\frac{\sqrt{(x+1)^2+4}}{2} + \frac{x+1}{2}\right| + C$$

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1. Evaluate the integrals.

4pts a) $\int t\,(\cos^2 t)\,dt$

$u = t \qquad dv = \cos^2 t\,dt$

$dv = \frac{1}{2} + \frac{1}{2}\cos 2t\,dt$

$du = dt \qquad v = \frac{1}{2}t + \frac{1}{4}\sin 2t$

$= t\left(\frac{1}{2}t + \frac{1}{4}\sin 2t\right) - \int\left(\frac{1}{2}t + \frac{1}{4}\sin 2t\right)dt$

$= \frac{1}{2}t^2 + \frac{1}{4}t\sin 2t - \left[\frac{1}{4}t^2 - \frac{1}{8}\cos 2t\right] + C$

$= \frac{1}{2}t^2 + \frac{1}{4}t\sin 2t - \frac{1}{4}t^2 + \frac{1}{8}\cos 2t + C$

$\frac{1}{4}t^2 + \frac{1}{4}t\sin 2t + \frac{1}{8}\cos 2t + C$

4pts b) $\int \sin^3(3x)\,dx = \int \sin^2 3x \sin 3x\,dx$

$\int (1-\cos^2 3x)\sin 3x\,dx \qquad u = \cos 3x$

$du = -3\sin 3x\,dx$

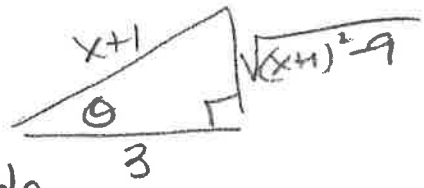
$\frac{du}{-3\sin 3x} = dx$

$-\frac{1}{3}\int(1-u^2)\,du$

$-\frac{1}{3}u + \frac{1}{9}u^3 + C$

$-\frac{1}{3}\cos 3x + \frac{1}{9}\cos^3 3x + C$

4pts e) $\int \frac{\sqrt{x^2+2x-8}}{x+1}\,dx$ $\quad \int \frac{\sqrt{(x+1)^2-9}}{x+1}\,dx$



$x+1 = 3\sec\theta$
$dx = 3\sec\theta\tan\theta\,d\theta$

$$\int \frac{\sqrt{9\sec^2\theta-9}}{3\sec\theta}(3\sec\theta\tan\theta\,d\theta) = \int \frac{3\tan\theta\; 3\cancel{\sec\theta}\tan\theta\,d\theta}{3\cancel{\sec\theta}}$$

$$3\int \tan^2\theta\,d\theta = 3\int(\sec^2\theta-1)\,d\theta = 3\tan\theta - 3\theta + C$$

$$3\left(\frac{\sqrt{(x+1)^2-9}}{3} - \sec^{-1}\left(\frac{x+1}{3}\right)\right) + C$$

4pts f) $\int \tan^3(2x)\,dx$ $\quad = \int \tan^2 2x \tan 2x\,dx$

$$\int(\sec^2 2x - 1)\tan 2x\,dx = \int \tan 2x \sec^2 2x\,dx - \int \tan 2x\,dx$$

$$\frac{1}{2}\,\frac{1}{2}(\tan^2 2x) - \frac{1}{2}\ln|\cos 2x| + C$$

$$\frac{1}{4}\tan^2(2x) - \frac{1}{2}\ln|\cos 2x| + C$$

5pts c) $\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}dx$

$x = 2\sin\theta \longrightarrow \begin{cases} 1 = 2\sin\theta & \theta = \pi/6 \\ \sqrt{3} = 2\sin\theta & \theta = \pi/3 \end{cases}$

$dx = 2\cos\theta\, d\theta$

$$\int_{\pi/6}^{\pi/3} \frac{\sqrt{4-4\sin^2\theta}}{4\sin^2\theta}(2\cos\theta\, d\theta) = \int_{\pi/6}^{\pi/3} \frac{2\cos\theta\, 2\cos\theta\, d\theta}{4\sin^2\theta}$$

$$\int_{\pi/6}^{\pi/3} \frac{\cos^2\theta}{\sin^2\theta}d\theta = \int_{\pi/6}^{\pi/3} \cot^2\theta\, d\theta = \int_{\pi/6}^{\pi/3} (\csc^2\theta - 1)d\theta$$

$$-\cot\theta - \theta \Big]_{\pi/6}^{\pi/3} = (-\cot\pi/3 - \pi/3) - (-\cot\pi/6 - \pi/6)$$

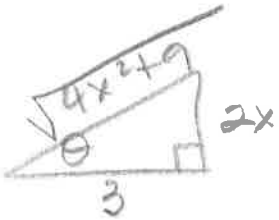
$$-\frac{\sqrt{3}}{3} - \frac{\pi}{3} + \sqrt{3} + \frac{\pi}{6} = \frac{2\sqrt{3}}{3} - \pi/6$$

4pts d) $\int \frac{1}{\sqrt{4x^2+9}}dx$

$2x = 3\tan\theta$

$x = 3/2 \tan\theta$

$dx = 3/2 \sec^2\theta\, d\theta$



$$\int \frac{1}{\sqrt{9\tan^2\theta+9}}(3/2\sec^2\theta\, d\theta) = \int \frac{3/2\sec^2\theta\, d\theta}{3\sec\theta}$$

$$\frac{1}{2}\int \sec\theta\, d\theta = \frac{1}{2}\ln|\sec\theta + \tan\theta| + C$$

$$\frac{1}{2}\ln\left|\frac{\sqrt{4x^2+9}}{3} + \frac{2x}{3}\right| + C$$